

1 Abstract

2 How species assemble temporally has received growing attention as
3 Climate change has shifted the timing of major life cycle events across most systems and is altering
4 how species assemble temporally. While theory predicts that plant species should partition time to
5 reduce resource competition—for example by leafing out at different times—few studies have examined
6 this given the complexity of environmental cues that make timings highly variable across years and
7 sites. We examine species- and community-level responses in budburst across 47 cuttings of woody
8 plant species across four sites in North America. By manipulating known temperature and photoperiod
9 cues in controlled environments, then applying a phylogenetic Bayesian model, we estimate the
10 effects of these cues and evolutionary history in shaping leafout across species. We found that each
11 cue caused budburst to shift at similar rates, despite plants being sampled at different sites, but that
12 the day of budburst varied greatly by species. These species-specific differences in their responses to
13 well-studied temperature and photoperiod cues, however, explained only half this variation, with the
14 remaining variation mostly explained by the estimated phylogenetic structure of budburst and inherent
15 species differences in the patterns and timing of their growth. This suggests current models of
16 leafout phenology are missing important variation that may structure species assembly within communities.
17 Identifying these unidentified cues or traits is critical to forecasting phenological shifts and
18 understanding their effects on community dynamics.

Introduction

Climate change has led to advances of about three days per decade in the timing of species life history events—phenology (Cohen et al., 2018; Parmesan, 2007; Thackeray et al., 2016). Different types of events, however, are highly variable, with certain events, such as leafout, spanning a period of weeks. Understanding and explaining this variability has become increasingly important as phenology shifts in response to rising temperatures at different rates (Fitter and Fitter, 2002; Fridley, 2012; Menzel et al., 2006; Parmesan, 2007; Yu et al., 2010). Some degree of variability is likely due to differences in climate change itself across space—as some areas warm faster than others and generate larger phenological shifts (Hoegh-Guldberg et al., 2018). Recent research suggests that species-level differences in phenology also drive some of this variation (Vitasse et al., 2009, 2018; Wolkovich and Cleland, 2014; Zohner and Renner, 2014). How these differences compare to other potential sources of variation, and what predicts them, is still largely unknown, but has major implications for predicting how changes in phenology will affect community dynamics and ecosystem services.

Differences in species phenologies can impact community structure and dynamics. Community ecology has long posited that species assemble in part through partitioning time, leading to unique temporal niches that allow species to coexist (Chesson et al., 2004; Grime, 1977). For example, earlier germination or leafout may provide understory species early-access to soil and light resources, before larger canopy species leafout (Heberling et al., 2019; Lee and Ibáñez, 2021; ?). These different temporal niches may also trade-off different pressures, with research suggesting that in seasonal systems species that are active earlier experience greater abiotic pressures—such as spring frost—but avoid the larger competitive pressures for resources that occur later in the season when most species are fully active (Augspurger, 2009; Gotelli and Graves, 1996; Pau et al., 2011; Sakai and Larcher, 1987). As climate change variably affects the timing of growth, it may alter the temporal order of species in a community and result in novel interactions between species (Cleland et al., 2012; Rudolf, 2019; Tiusanen et al., 2020; ?), thus changing the competitive landscape and synchrony of interactions within communities.

While recent changes in climate are shaping differences in phenology, for many species their temporal niches may also reflect their evolutionary history. Species traits in communities today are the result of selective pressures acting over evolutionary timescales that include the historical climates and related ancestral phenotypes that were favourable under previous conditions (Wiens et al., 2010). This could produce greater similarity in phenological traits of more closely related species (Davies et al., 2013; Morales-Castilla et al., 2024), which would impact how species respond to future climates.

53 Spring budburst offers an excellent system to test for species- and community-level patterns in phe-
54 nology and environmental cues. Budburst of temperate woody plants is well studied and thus known
55 to respond to environmental cues, especially temperatures in the winter and spring—referred to as
56 chilling and forcing, respectively—and daylength or photoperiod (Basler and Körner, 2014; Chuine
57 et al., 2010; Cooke et al., 2012; Laube et al., 2014; Polgar and Primack, 2011). Budburst responses
58 to these cues are generally similar across sites and populations (Aitken and Bemmels, 2016; Zeng and
59 Wolkovich, 2024). Species-level differences, however, are generally much larger than observations of
60 site or population-level variation (but see Deans, 1996; Sogaard et al., 2008, for examples of population-
61 level differences). Decades of phenological research suggests these unique phenologies are likely due to
62 different responses of species to temperature and photoperiod cues, though they may be also structured
63 by species shared evolutionary history (Davies et al., 2013; Gougherty and Gougherty, 2018; Lechowicz,
64 1984) and their current environments. For example, research predicts that forest understory species
65 should vary their leafout and budburst early to both gain access to light and soil nutrients prior to
66 canopy closure (Mahall and Bormann, 1978; Muller, 1978), and to reduce competition between species
67 within the understory versus canopy.

68
69 Here we test how environmental—temperature and photoperiod—cues and evolutionary relationships
70 shape species-level variation in phenology of forest communities across North America. We leveraged
71 two large-scale growth chamber cutting experiments, in which we vary the temperature cues of chilling
72 (cool temperatures in the fall and winter) and forcing (warmer temperatures, usually in the spring) with
73 photoperiod (Fig. 1), with Bayesian phylogenetic approach to estimate budburst responses across 47
74 woody species. We collected samples from plants at four sites, in eastern and western North America
75 (Fig. 1), with pairs of sites on each coast spanning 4 and 6° latitude, respectively. Our dataset
76 includes the diverse assemblages of tree and shrub species found in eastern and western deciduous
77 forests, allowing us to explore differences across both sites and functional groups—including understory
78 shrubs versus canopy trees—in eastern versus western forests.

79 **Materials and Methods**

80 **Field sampling**

81 We combined data from two growth chamber studies using dormant branch clippings of North Ameri-
82 can deciduous woody plants. Such cutting experiments are a common and powerful approach to infer
83 the responses of adult trees to environmental conditions (Vitasse et al., 2014; ?). In our first study,
84 we collected samples from two eastern communities—Harvard Forest, Massachusetts, USA (42.55°N,

72.20°W) and St. Hippolyte, Quebec, Canada (45.98°N, 74.01°W), from 20-28 January, 2015 (previously reported in Flynn and Wolkovich, 2018). The second growth chamber study spanned two western communities—Smithers (54.78°N, 127.17°W) and E.C. Manning Park (49.06°N, 120.78°W), British Columbia, Canada, with sampling from 19-28 October 2019 (not previously published). Combining the datasets from these two experiments allowed us to test for spatial variation in the phenology of woody plant communities at a larger geographic scale and make stronger inferences for how different species vary in their cues.

We selected the dominant deciduous species in each forest community and maximized the number of species occurring across sites. We observed budburst for 47 species spanning eastern and western forest communities, with 28 species at our eastern transect and 21 species at our western transect, of which 3 species occurred at both regions (Fig. 1, Table S1). Of the species in our eastern transect, there were 13 shrubs and 15 trees, while our western transect included 18 shrub and 4 tree species. At all sites, we tagged 15-20 healthy, individuals for each species in the summer prior to sampling. To reduce ontogenetic effects (Vitasse et al., 2014), we only sampled mature individuals, which we assumed to be trees that were greater than 2 m in height and with DBH greater than 4 cm (Natural Resources Canada, 2020). We collected a single cutting from small shrub species, but depending on the size of the sampled tree, we took between 6-16 cuttings, thus collecting samples from 6-20 tagged plants per species. All cuttings were taken from the same relative canopy position, sampling terminal branches from shrubs and the lower canopy from trees using a pole pruner from the ground. We kept samples cold and immediately placed them in erlenmeyer flasks of water upon returning from the field. We conducted the eastern study at the Arnold Arboretum, Boston, MA, USA, and the western study at the University of British Columbia, BC, Canada. We used daily weather data from local weather stations to calculate the field chilling prior to collection (see Table S2).

Growth chamber study

We used growth chambers with two levels of chilling—with either no additional chilling or 30 days at 4°C for our eastern study, and 21 or 56 days of chilling at 4°C for our western study (all dark)—after which we moved plants to one of two levels of forcing—with day and night temperatures that varied to create a cool regime of 15:5°C and a warm regime of 20:10°C—crossed with two photoperiods of either 8 or 12 hours—for a total of 8 distinct treatments in each study.

Our design was similar for both eastern and western sites (Fig. 1). We had seven to ten replicates per treatment in our eastern study and eight replicates in our western study, with the exception of *Sorbus*

118 *scopulina*, for which we only had six replicates per treatment. Our eastern and western studies differed
 119 only in the timing of sample collection and thermoperiodicity (the interval of alternating day and
 120 night temperatures) in forcing treatments. By collecting samples in our eastern study in late January,
 121 they experienced considerable field chilling. We collected the western samples in late October, with
 122 most chilling occurring under controlled conditions. In our eastern study, we also set the duration of
 123 forcing temperatures to correspond with the photoperiod treatments. While this is a widespread and
 124 common approach in phenology chamber studies, it leads to a confounding effect in the thermoperi-
 125 odicity (Buonaiuto et al., 2023). Thus we applied forcing temperatures for 12 hours independent of
 126 the duration of the photoperiod treatments in our western study, and controlled for this difference in
 127 our statistical approach (see below). Each experiment was conducted across 8 growth chambers. We
 128 additionally rotated cuttings within chambers approximately every two weeks to avoid any chamber
 129 effects and changed the water and re-cut each cutting to prevent callusing and keep cuttings healthy.
 130 We controlled for small differences in the number of replicates across species in our statistical approach
 131 using partial pooling (see section on statistical analysis).

132
 133 We assessed phenological stages of budburst using the BBCH scale (Finn et al., 2007), adapted for our
 134 specific species. We also created photographic guides to help define each stage of the BBCH scale for
 135 our specific species (Loughnan and Wolkovich, 2024; Savas et al., 2017). Every 1-3 days, we observed
 136 each sample and recorded all phenological stages up to full leaf expansion (defined as code 17 by Finn
 137 et al., 2007). Here, we modelled the timing of budburst, defined as the day of budbreak or shoot
 138 elongation (defined as code 07 by Finn et al., 2007) since forcing treatments began. Our eastern study
 139 spanned 82 days, with over 19320 phenological observations, while the western study spanned 113 days
 140 and 47844 phenological observations.

142 Statistical Analysis

143 We tested for differences in budburst cues across species and communities using a phylogenetic mixed
 144 effects model with partial pooling (‘shrinkage’) across species. This approach accounts for both the
 145 evolutionary relatedness of our species and estimates the species-level cues and differences across sites.
 146 We obtained species phylogenetic relatedness by pruning the Smith and Brown (2018) megatree of
 147 angiosperms (Fig. 2).

148
 149 In our model, we included each of our three cues as continuous variables. To account for the greater
 150 amounts of field chilling experienced in our eastern study, we converted our chilling treatments to chill

portions for each individual community using local weather station data (Table S2) and the chillR package (v. 0.73.1, Luedeling, 2020). Chill portions are a commonly used chill metric that is calculated using hourly climate data and a Dynamic Model in which cool temperatures lead to the accumulation of chilling, while more accurately estimating chilling accumulation under fluctuating temperatures (Luedeling, 2012, 2020). We calculated chill portions for both the chilling experienced in the field prior to sampling and during the experiment in our 4°C chilling chambers. We also converted forcing temperatures to mean daily temperatures in each treatment to account for differences in thermoperiodicity between the two studies (Buonaiuto et al., 2023). Finally, we z -scored each cue and site using two standard deviations to allow direct comparisons between results across cues (Gelman, 2008). In addition to the overall predictors, we included all two-way interactions between cues and between cues and sites.

Our modelling approach allowed us to combine observations of budburst (y_i) across species (sp) to estimate both the effects of species-specific differences in their responses (the rate of change in budburst day per unit of standardized cue, as estimated by the slope, β) to cues (chilling, forcing and photoperiod) and any remaining species-specific differences in budburst day (via species-level intercepts, α_{sp_i}). Our full model also estimated effects of sites and all possible two-way interactions between cues, and between cues and sites, to estimate the day of budburst ($\hat{y}$) relative to the first day of forcing conditions and relative to baseline conditions and our Smithers site.

$$\begin{aligned}\hat{y}_i = & \alpha_{sp_i} + \beta_{Manning\ park} + \beta_{Harvard\ forest} + \beta_{St.\ Hippolyte} + \\ & \beta_{chilling_{sp_i}} + \beta_{forcing_{sp_i}} + \beta_{photoperiod_{sp_i}} + \\ & \beta_{forcing \times chilling_{sp_i}} + \beta_{forcing \times photoperiod_{sp_i}} + \beta_{chilling \times photoperiod_{sp_i}} + \\ & \beta_{forcing \times Manning\ park_{sp_i}} + \beta_{forcing \times Harvard\ forest_{sp_i}} + \beta_{forcing \times St.\ Hippolyte_{sp_i}} + \\ & \beta_{chilling \times Manning\ park_{sp_i}} + \beta_{chilling \times Harvard\ forest_{sp_i}} + \beta_{chilling \times St.\ Hippolyte_{sp_i}} + \\ & \beta_{photoperiod \times Manning\ park_{sp_i}} + \beta_{photoperiod \times Harvard\ forest_{sp_i}} + \beta_{photoperiod \times St.\ Hippolyte_{sp_i}}\end{aligned}$$

$$y_i \sim \text{normal}(\hat{y}_i, \sigma^2)$$

The slopes were modelled with partial pooling at the species-level:

$$\begin{aligned}\beta_{forcing_{sp}} & \sim \text{normal}(\mu_{forcing_{sp}}, \sigma_{forcing}^2) \\ \dots & \\ \beta_{photoperiod \times St.Hippolyte_{sp}} & \sim \text{normal}(\mu_{photoperiod \times St.Hippolyte_{sp}}, \sigma_{photoperiod \times St.Hippolyte_{sp}}^2)\end{aligned}$$

We included the phylogenetic effect on species-specific differences in budburst timing (α_{sp}) as a variance covariance matrix ($\mathbf{V}$) in the parameterization of the normal random vector:

$$\boldsymbol{\alpha}_{sp} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \dots \\ \alpha_n \end{bmatrix} \text{ such that } \boldsymbol{\alpha} \sim \text{multi-normal}(\mu_{\alpha}, \mathbf{V})$$

The timing of budburst ancestrally is represented as the root value of the means of the multivariate normal distribution. The phylogenetic structuring of budburst is estimated using Pagel’s lambda, a phylogenetic scaling parameter where values close to one indicate a phylogenetic signal consistent with Brownian motion and values close to zero the absence of a phylogenetic signal for a given trait (Pagel, 1999; Pearse et al., 2025). For more detail on this method of phylogenetic modeling, see Morales-Castilla et al. (2024).

We used non-informative priors for each of our model parameters (increasing priors three-fold did not demonstrably change our estimates), and validated the model code using test data prior to analysis. Our model produced $\hat{R}$ values close to 1, well mixed chains, and n_{eff} values that exceeded 10% of the model iterations. Our model was fit using the Stan language with the rstan package (v. 2.26.23, Stan Development Team, 2018) in R (v. 4.3.1, R Development Core Team, 2017).

We present model estimates, relative to baseline or low treatment conditions. This means that we show the average daily temperature for our cooler forcing conditions (which averaged to 8.33°C for our eastern sites and 10°C for our western sites, after accounting for the differences in thermoperiodicity), with a short photoperiod (8 hours photoperiod), and the mean of lowest chill portions from each population (which is equivalent to approximately 10 weeks of chilling at 4°C) and—when relevant—our most northern sites. We present results as means and 90% posterior uncertainty intervals from our Bayesian phylogenetic models.

Results

On average, species budburst occurred 30.4 days (uncertainty interval: 26.0, 35.2) after the start of forcing and photoperiod treatments (all estimates are given as mean $\pm$ 90% uncertainty intervals, henceforth ‘UI,’ and budburst dates given as relative to baseline conditions, see methods for more details). Between the earliest and latest budbursting species, the difference in the mean day of budburst

was 38.5 and 30.7 days for the eastern and western sites respectively. Budburst of individual species showed distinct differences in their timing and the relative importance of cues, but all cues led to an advance in budburst date. Of the three cues, chilling had the largest effect, with increased chilling advancing budburst (we represent advances with negative numbers, while delays are positive; -14.8, UI: -16.7, -12.9 days per standardized chill portions) and photoperiod the smallest (-3.4, UI: -4.0, -2.7 days per standardized photoperiod, Fig. 3a, S1). But we found a large, subadditive interaction between forcing and chilling (8.4, UI: 7.0, 9.8, Fig. 3b); meaning that low chilling is offset by high forcing conditions, and vice versa (Table S3).

Overall we found species budburst was moderately phylogenetically structured (Pagel's λ of 0.4, UI: 0.1, 0.8, Fig 2, estimated on the intercept). This phylogenetic structure suggests latent processes that shape species-level differences in budburst timing beyond the species-specific differences in their shifts responses to temperature (chilling and forcing) and photoperiod. The contribution of these species-specific cue differences to budburst order and timing was small compared to the overall species differences unexplained by cues (represented in the modeling framework as the intercept), with the exact magnitude for which chilling, forcing and photoperiod determine budburst date depending on the level of each cue (e.g., forcing has a much larger effect on budburst timing given a constant day-time temperature of 20°C compared to a constant daytime temperature of 15°C). Across the levels of chilling, forcing and photoperiod we studied, species-specific differences in responses to cues explained only between 37.6% and 67.8% of variation for eastern species and between 47.4 and 61.7% for western species, with the phylogenetically structured intercept explaining the rest (Fig. 2).

We also found the rank order of species' budburst timings changed more when estimated using higher chilling (the average of which is equivalent to approximately 15 weeks chilling at 4°C), forcing and longer photoperiods than under lower chilling and forcing and shorter photoperiods (Fig. 1,S2). The budburst ranking for the three species that occurred in both transects remained relatively consistent across the two communities (Fig. 4), with only *Alnus incana* in the western community experiencing a large change in rank with cues (Fig. S2). In comparing the earliest and latest budbursting species, we found relatively small differences in cues (Fig. 4). For example, an early budbursting shrub, *Lyonia ligustrina*, had chilling and forcing cue estimates of -14.7 and -11.5 respectively, which are comparable to the cue estimates of -15.7 and -11.5 for the much later tree, *Quercus alba*. Yet the model predicts these species to have a 15.1 day difference in day of budburst, highlighting how the timing of budburst across species was also strongly related to species level differences (intercept values) independent of their responses to chilling, forcing and photoperiod (see Fig. 5).

Despite our expectations that species responses should differ across the canopy layer, with shrubs being less sensitive to cues than trees, we did not find substantial differences across these two functional groups (Fig. S3). Shrubs, like *Cornus stolonifera*, budburst early and showed small responses to chilling and forcing (Fig. 4b, d). But 36.7% of the shrubs, including *Menziesia ferruginea* and *Symphoricarpos alba*, exhibited the opposite response and budburst relatively late (Fig. 5, S3). Similarly for trees, some species did match our predictions, exhibiting stronger cues, but 23.5% of species budburst earlier than expected. While specific tree species, such as *Quercus velutina*, did have larger chilling and photoperiod responses as predicted (Fig. 4a, e), and *Fagus grandifolia* had the largest photoperiod response (Fig. 4e), overall we did not find clear differences between the cues of trees and shrubs across the four forest communities.

Finally, we observed negligible site-level effects compared to differences between cues (Fig. 3, Table S3). Treatments that received a month less of chilling (low chilling), were 5°C cooler (low forcing), and had shorter photoperiods all budburst later compared to the more chilled, warmer and long photoperiod treatments across all sites and across transects (Fig. S1 a-c, shown for baseline conditions, see Table S3 for model output). Overall budburst dates did not differ between sites, though eastern sites budburst marginally earlier (day 35.8, UI: 28.3, 42.3) compared to the western sites (day 48.1, UI: 40.7, 54.5, see also Table S3 for model output). This could be due to the earlier collection date of our western species, which reduced the field chilling they received relative to our eastern samples (though see corrections for this in the statistical analysis section of our methods).

Discussion

To accurately forecast shifts in phenology and its impact on ecological communities requires understanding which environmental cues are most important at the population-, species-, functional-group and community-levels. Our results support decades of research finding that budburst timing advances in response to the well-studied major temperature cues of chilling and forcing (with the greatest responses to chilling and forcing respectively, see also Ettinger et al., 2020; Flynn and Wolkovich, 2018) and longer photoperiods. But using our modelling approach allowed us to estimate the extent to which these species-specific responses to these cues do—or do not—lead to differences in species timings that may structure temporal niches within communities. Perhaps surprisingly, we found a large portion (one-third to one-half) of variation in species budburst timings was unexplained by different responses to chilling, forcing and photoperiod. Our results show that some of this variation is due to species evolutionary history, but not all, suggesting fundamental gaps in our model of one of the best studied

and most important phenological events—woody plant budburst.

Differences in species responses to temperature and photoperiod were similar across sites and functional groups, suggesting that understanding the drivers of variation across species may be an important task to improve forecasting. Our study included 47 of the dominant woody species from deciduous forests across North America from both shrub and tree functional groups, with a mix of early and late budbursting phenologies. Despite the high degree of diversity and spatial scale represented in our study, we found no detectable site-level differences, including across different coasts and over 6° of latitude. This contrasts with research that suggests latitudinal trends in photoperiod and temperature may impose unique selective pressures, and thus may drive site-level differences in responses to cues (Keller et al., 2011). Latitudinal gradients in spring phenological shifts have been found by studies using *in situ* phenological data (Alecrim et al., 2023; Post et al., 2018). Such work, however, is based on long-term observations in the field, with confounding differences in study duration and start dates, variable methodologies, and geographic extent. The lack of site-level effects that we observed—using experiments to estimate responses—suggests budburst cues are not shaped by local conditions, at least at our study scale.

Community composition and interspecific variation in phenology

The variation across species in their responses to environmental cues has the potential to create large differences in species temporal niches and ecological roles. Species ranged from early to late budburst timing in our experimental conditions, spanning a similar period as observed in natural communities (Lechowicz, 1984; Richardson, A.D., O’Keefe, 2009). This suggests our experiment captured a realistic breadth in temperate forest budburst. Furthermore, all our focal species responded to each environmental cue, with large responses to chilling and comparatively small responses to photoperiod—trends consistent with previous studies (Ettinger et al., 2020; Flynn and Wolkovich, 2018). Cues, however, had complex interactions that appear potentially advantageous under warming climates (Caffarra and Donnelly, 2011; Flynn and Wolkovich, 2018; Heide, 1993). The interaction between chilling and forcing may ensure that species budburst even if warmer winters cause insufficient chilling, but will require additional forcing. But despite these differences in species responses to cues, we did not find clear, generalizable trends across functional groups.

Shrubs and trees differ greatly in their physiology, filling different ecological niches. Most of our trees budburst later than our study’s shrubs, a relative order also found by previous studies (Gill et al., 1998; Panchen et al., 2014). But about a quarter of our trees also budburst early, with timing more similar

to that of shrubs. Similarly, we found a third of shrubs budburst at similar times as the majority of our trees. These deviations from our expectations for how shrubs and trees partition their budburst timing suggests there is more nuance to these patterns. As tree species advance phenologically—resulting in earlier canopy closure and reduced light availability (Donnelly and Yu, 2019; ?)—this subset of shrubs with small cues may have reduced fitness. Our findings suggest, however, that many shrubs will also advance phenologically and have the potential to maintain their relative temporal niche.

Community assembly in responses to cues versus evolutionary history

Our study included the three cues—chilling, forcing and photoperiod—commonly thought to determine budburst timing, but over a third of the total variation was not explained by these cues (Fig. 4), suggesting the general model of budburst may be incomplete. This, paired with the observed phylogenetic structure in budburst timing, hints at unidentified latent traits still missing from our understanding of the processes shaping spring phenology (Davies et al., 2019; Webb et al., 2002).

Latent traits could reflect other cue or trait relationships indicative of species growth strategies or key traits that may facilitate greater resource use, or the greater competitive abilities needed when budbursting later (Grime, 1977). For example, species that budburst later in the season may have more conservative growth strategies, possessing traits that allow them to compete for limited soil resources and light, such as low specific leaf areas or higher wood densities (Chave et al., 2009; Wright et al., 2004). But our results suggest that evolutionary history and species ancestral phenotype also have some effect on species timings and cues. Species traits are shaped over ‘deep’ timescales, with previous evolution also influencing responses to the present climate. Phenological trends could thus be an indication of limited evolutionary time to adapt (Lechowicz, 1984), or shaped by selection on other important traits that are not phylogenetically structured. Incorporating both phenology and evolutionary history into a broader trait framework could thus provide insights into traits that correlate with budburst timing, and—ultimately—the current and evolutionary drivers that select for species phenotypes and possibly shape species adaptive potential to future climates.

Predicting budburst under future climates

As climate change leads to greater temperature cues, we are already observing advances in species budburst within diverse ecological communities. The greatest ecological impacts will occur in high latitude communities that are warming the most and the fastest (Hoegh-Guldberg et al., 2018), mak-

ing species in northern communities most likely to advance their budburst and experience the greatest changes in budburst order. Yet our ability to predict these changes is often limited by the lack of data available. Our findings may provide insights into phenological cues in other temperate forests or communities that experience similar temperature regimes, but may underestimate responses in other plants functional groups, such as herbaceous species, that are known to exhibit more variability in their phenology (Horbach et al., 2023). The consistent trends we observed across sites would still support forecasting across similar regions with limited local phenological data but similar species assemblages, allowing us to leverage existing research to mitigate the cascading effects that shifted plant phenology could have on ecological communities.

While our findings have applications to plant communities, our analytical approach is widely applicable to forecasting diverse species responses to climate change. The phylogenetic model we used here could be easily applied to other phenological events or suites of species from across the tree of life. By combining species-rich and community-wide approaches with phylogenetic relationships we can better understand the impacts of evolutionary history and current ecological processes in shaping species phenology under a changing climate.

Acknowledgements

We thank M. Yasutake, K. Slimon, S. Larter, P. Autio, and S. Collins for their help running the Western part of this experiment; D. Flynn and T. Savas for contributions to the Eastern experimental data; S. Joly for his help constructing the phylogenetic tree used in our analysis; and to J. Hopps and K. Wilson for their support during our work at E. C. Manning Park and P. Crawford and D. Belford for their support while working in Babine Provincial Park.

Author Contribution Statement

Both D.L. and E.M.W. conceived the study, contributed to the analysis and code and contributed to the writing and revision of the manuscript. DL led field and experimental data collection.

Competing Interests Statement

The authors have no competing interests.

References

- Aitken, S. N., and J. B. Bemmels. 2016. Time to get moving: assisted gene flow of forest trees. *Evolutionary Applications* **9**:271–290.
- Alecrim, E. F., R. D. Sargent, and J. R. K. Forrester. 2023. Higher-latitude spring-flowering herbs advance their phenology more than trees with warming temperatures. *Journal of Ecology* **111**:156–169.
- Augspurger, C. K. 2009. Spring 2007 warmth and frost : phenology , damage and refoliation in a temperate deciduous forest. *Functional Ecology* **23**:1031–1039.
- Basler, D., and C. Körner. 2014. Photoperiod and temperature responses of bud swelling and bud burst in four temperate forest tree species. *Tree Physiology* **34**:377–388.
- Buonaiuto, D. M., E. M. Wolkovich, and M. J. Donahue. 2023. Experimental designs for testing the interactive effects of temperature and light in ecology : The problem of periodicity. *Functional Ecology* **37**:1747–1756.
- Caffarra, A., and A. Donnelly. 2011. The ecological significance of phenology in four different tree species: Effects of light and temperature on bud burst. *International Journal of Biometeorology* **55**:711–721.
- Chave, J., D. Coomes, S. Jansen, S. L. Lewis, N. G. Swenson, and A. E. Zanne. 2009. Towards a worldwide wood economics spectrum. *Ecology Letters* **12**:351–366.
- Chesson, P., R. L. E. Gebauer, S. Schwinning, N. Huntly, K. Wiegand, M. S. K. Ernest, A. Sher, A. Novoplansky, and J. F. Weltzin. 2004. Resource pulses, species interactions, and diversity maintenance in arid and semi-arid environments. *Oecologia* **141**:236–253.
- Chuine, I., X. Morin, and H. Bugmann. 2010. Warming, photoperiods, and tree phenology. *Science* **329**:277–278.
- Cleland, E. E., J. M. Allen, T. M. Crimmins, J. A. Dunne, S. Pau, S. E. Travers, E. S. Zavaleta, and E. M. Wolkovich. 2012. Phenological tracking enables positive species responses to climate change. *Ecology* **93**:1765–1771.
- Cohen, J. M., M. J. Lajeunesse, and J. R. Rohr. 2018. A global synthesis of animal phenological responses to climate change. *Nature Climate Change* **8**:224–228.
- Cooke, J. E., M. E. Eriksson, and O. Junttila. 2012. The dynamic nature of bud dormancy in trees: Environmental control and molecular mechanisms. *Plant, Cell and Environment* **35**:1707–1728.

394 Davies, T. J., J. Regetz, E. M. Wolkovich, and B. J. McGill. 2019. Phylogenetically weighted regression:
395 A method for modelling non-stationarity on evolutionary trees. *Global Ecology and Biogeography*
396 **28**:275–285.

397 Davies, T. J., E. M. Wolkovich, N. J. B. Kraft, N. Salamin, J. M. Allen, T. R. Ault, J. L. Betancourt,
398 K. Bolmgren, E. E. Cleland, B. I. Cook, T. M. Crimmins, S. J. Mazer, G. J. McCabe, S. Pau,
399 J. Regetz, M. D. Schwartz, and S. E. Travers. 2013. Phylogenetic conservatism in plant phenology.
400 *Journal of Ecology* **101**:1520–1530.

401 Deans, J. 1996. Frost hardiness of 16 European provenances of sessile oak growing in Scotland. *Forestry*
402 **69**:5–12.

403 Donnelly, A., and R. Yu. 2019. Temperate deciduous shrub phenology: the overlooked forest layer.
404 *International Journal of Biometeorology* **65**:343–355.

405 Ettinger, A. K., C. J. Chamberlain, I. Morales-Castilla, D. M. Buonaiuto, D. F. Flynn, T. Savas,
406 J. A. Samaha, and E. M. Wolkovich. 2020. Winter temperatures predominate in spring phenological
407 responses to warming. *Nature Climate Change* **10**:1137–1142.

408 Finn, G. A., A. E. Straszewski, and V. Peterson. 2007. A general growth stage key for describing trees
409 and woody plants. *Annals of Applied Biology* **151**:127–131.

410 Fitter, A. H., and R. S. Fitter. 2002. Rapid changes in flowering time in British plants. *Science*
411 **296**:1689–1691.

412 Flynn, D. F. B., and E. M. Wolkovich. 2018. Temperature and photoperiod drive spring phenology
413 across all species in a temperate forest community. *New Phytologist* **219**:1353–1362.

414 Fridley, J. D. 2012. Extended leaf phenology and the autumn niche in deciduous forest invasions.
415 *Nature* **485**:359–362.

416 Gelman, A. 2008. Scaling regression inputs by dividing by two standard deviations. *Statistics in*
417 *Medicine* **27**:2865–2873.

418 Gill, D. S., J. S. Amthor, and F. H. Bormann. 1998. Leaf phenology, photosynthesis, and the persistence
419 of saplings and shrubs in a mature northern hardwood forest. *Tree Physiology* **18**:281–289.

420 Gotelli, N. J., and G. R. Graves, 1996. The temporal niche. Pages 95–112 *in* *Null Models In Ecology*.
421 Smithsonian Institution Press, Washington, D. C.

422 Gougherty, A. V., and S. W. Gougherty. 2018. Sequence of flower and leaf emergence in deciduous
423 trees is linked to ecological traits, phylogenetics, and climate. *New Phytologist* **220**:121–131.

424 Grime, J. P. 1977. Evidence for the Existence of Three Primary Strategies in Plants and Its Relevance
425 to Ecological and Evolutionary Theory. *The American Naturalist* **111**:1169–1194.

426 Heberling, J. M., S. T. Cassidy, J. D. Fridley, and S. Kalisz. 2019. Carbon gain phenologies of spring-
427 flowering perennials in a deciduous forest indicate a novel niche for a widespread invader. *New*
428 *Phytologist* **221**:778–788.

429 Heide, O. M. 1993. Daylength and thermal time responses of budburst during dormancy release in
430 some northern deciduous trees. *Physiologia Plantarum* **88**:531–540.

431 Hoegh-Guldberg, O., D. Jacob, M. Taylor, M. Bindi, S. Brown, I. Camilloni, A. Diedhiou, R. Djalante,
432 K. Ebi, F. Engelbrecht, J. Guiot, Y. Hijioka, S. Mehrotra, A. Payne, S. Seneviratne, A. Thomas,
433 R. Warren, and G. Zhou, 2018. Impacts of 1.5°C Global Warming on Natural and Human Systems.
434 In: *Global Warming of 1.5 °C. An IPCC Special Report on the impacts of global warming of 1.5 °C*
435 *above pre-industrial levels and related global greenhouse gas emission pathways, in the context of .*
436 *Technical report, Cambridge University Press, Cambridge, UK and New York, NY, USA.*

437 Horbach, S., R. Rauschkolb, and C. Römermann. 2023. Flowering and leaf phenology are more variable
438 and stronger associated to functional traits in herbaceous compared to tree species. *Flora* **300**:1–9.

439 Keller, S. R., R. Y. Soolanayakanahally, R. D. Guy, S. N. Silim, M. S. Olson, and P. Tiffin. 2011.
440 Climate-driven local adaptation of ecophysiology and phenology in balsam poplar, *Populus balsam-*
441 *ifera* L. (Salicaceae). *American Journal of Botany* **98**:99–108.

442 Laube, J., T. H. Sparks, N. Estrella, J. Höfler, D. P. Ankerst, and A. Menzel. 2014. Chilling outweighs
443 photoperiod in preventing precocious spring development. *Global Change Biology* **20**:170–182.

444 Lechowicz, M. J. 1984. Why Do Temperate Deciduous Trees Leaf Out at Different Times? *Adaptation*
445 *and Ecology of Forest Communities. The American Naturalist* **124**:821–842.

446 Lee, B. R., and I. Ibáñez. 2021. Spring phenological escape is critical for the survival of temperate
447 tree seedlings. *Functional Ecology* **35**:1848–1861.

448 Loughnan, D., and E. M. Wolkovich. 2024. A photographic guide to spring phenology for woody plants
449 in Western North America. *Knowledge Network for Biocomplexity (KNB)* .

450 Luedeling, E. 2012. Climate change impacts on winter chill for temperate fruit and nut production:
451 A review. *Scientia Horticulturae* **144**:218–229.

452 Luedeling, E., 2020. chillR: Statistical Methods for Phenology Analysis in Temperate Fruit Trees.
453 <https://CRAN.R-project.org/package=chillR>.

454 Mahall, B. E., and F. H. Bormann. 1978. A Quantitative Description of the Vegetative Phenology of
 455 Herbs in a Northern Hardwood Forest. *Botanical Gazette* **139**:467–481.

456 Menzel, A., T. H. Sparks, N. Estrella, E. Koch, A. Aaasa, R. Ahas, K. Alm-Kübler, P. Bissolli,
 457 O. Braslavská, A. Briede, F. M. Chmielewski, Z. Crepinsek, Y. Curnel, Å. Dahl, C. Defila, A. Don-
 458 nelly, Y. Filella, K. Jactzak, F. Måge, A. Mestre, Ø. Nordli, J. Peñuelas, P. Pirinen, V. Remišová,
 459 H. Scheifinger, M. Striz, A. Susnik, A. J. Van Vliet, F. E. Wielgolaski, S. Zach, and A. Zust. 2006.
 460 European phenological response to climate change matches the warming pattern. *Global Change*
 461 *Biology* **12**:1969–1976.

462 Morales-Castilla, I., T. J. Davies, G. Legault, D. M. Buonaiuto, C. J. Chamberlain, A. K. Ettinger,
 463 M. Garner, F. A. M. Jones, D. Loughnan, W. D. Pearse, D. S. Sodhi, and E. M. Wolkovich. 2024.
 464 Phylogenetic estimates of species-level phenology improve ecological forecasting. *Nature Climate*
 465 *Change* **14**:989–995.

466 Muller, R. N. 1978. The Phenology, Growth and Ecosystem Dynamics of *Erythronium americanum*
 467 in the Northern Hardwood Forest. *Ecological Monographs* **48**:1–20.

468 Natural Resources Canada, 2020. Forestry glossary: Sapling. URL [https://cfs.nrcan.gc.ca/terms/](https://cfs.nrcan.gc.ca/terms/read/996)
 469 [read/996](https://cfs.nrcan.gc.ca/terms/read/996).

470 Pagel, M. 1999. Inferring the historical patterns of biological evolution. *Nature* **401**:877–884.

471 Panchen, Z. A., R. B. Primack, B. Nordt, E. R. Ellwood, A. Stevens, S. S. Renner, C. G. Willis,
 472 R. Fahey, A. Whittemore, Y. Du, and C. C. Davis. 2014. Leaf out times of temperate woody
 473 plants are related to phylogeny, deciduousness, growth habit and wood anatomy. *New Phytologist*
 474 **203**:1208–1219.

475 Parmesan, C. 2007. Influences of species, latitudes and methodologies on estimates of phenological
 476 response to global warming. *Global Change Biology* **13**:1860–1872.

477 Pau, S., E. M. Wolkovich, B. I. Cook, T. J. Davies, N. J. B. Kraft, K. Bolmgren, J. L. Betancourt, and
 478 E. E. Cleland. 2011. Predicting phenology by integrating ecology, evolution and climate science.
 479 *Global Change Biology* **17**:3633–3643.

480 Pearse, W. D., T. J. Davies, and E. M. Wolkovich. 2025. How to define, use, and Interpret Pagel’s
 481 lambda in Ecology and Evolution. *Global Ecology and Biogeography* **34**:e70012.

482 Polgar, C. A., and R. B. Primack. 2011. Leaf-out phenology of temperate woody plants: From trees
 483 to ecosystems. *New Phytologist* **191**:926–941.

Post, E., B. A. Steinman, and M. E. Mann. 2018. Acceleration of phenological advance and warming with latitude over the past century. *Scientific Reports* **8**:1–8.

R Development Core Team, 2017. R: A Language and Environment for Statistical Computing.

Richardson, A.D., O’Keefe, J., 2009. Phenological Differences Between Understory and Overstory. Pages 87–117 in A. Noormets, editor. *Phenology of Ecosystem Processes*. Springer US, New York, NY.

Rudolf, V. H. W. 2019. The role of seasonal timing and phenological shifts for species coexistence. *Ecology Letters* **22**:1324–1338.

Sakai, A., and W. Larcher. 1987. *Frost Survival of Plants: Responses and adaptation to freezing stress*. Springer-Verlag, Berlin, Heidelberg.

Savas, T., D. F. B. Flynn, and E. M. Wolkovich. 2017. A standardized photographic guide to woody plant spring phenology. Knowledge Network for Biocomplexity (KNB) .

Smith, S. A., and J. W. Brown. 2018. Constructing a broadly inclusive seed plant phylogeny. *American Journal of Botany* **105**:302–314.

Sogaard, G., O. Johnsen, J. Nilsen, and O. Junttila. 2008. Climatic control of bud burst in young seedlings of nine provenances of Norway spruce. *Tree Physiology* **28**:311–320.

Stan Development Team, 2018. RStan: the R interface to Stan. URL <http://mc-stan.org/>.

Thackeray, S. J., P. A. Henrys, D. Hemming, J. R. Bell, M. S. Botham, S. Burthe, P. Helaouet, D. G. Johns, I. D. Jones, D. I. Leech, E. B. MacKay, D. Massimino, S. Atkinson, P. J. Bacon, T. M. Brereton, L. Carvalho, T. H. Clutton-Brock, C. Duck, M. Edwards, J. M. Elliott, S. J. Hall, R. Harrington, J. W. Pearce-Higgins, T. T. Høye, L. E. Kruuk, J. M. Pemberton, T. H. Sparks, P. M. Thompson, I. White, I. J. Winfield, and S. Wanless. 2016. Phenological sensitivity to climate across taxa and trophic levels. *Nature* **535**:241–245.

Tiusanen, M., T. Kankaanpää, N. S. Schmidt, and T. Roslin. 2020. Heated rivalries: Phenological variation modifies competition for pollinators among arctic plants. *Global Change Biology* **26**:6313–6325.

Vitasse, Y., D. Basler, and D. Way. 2014. Is the use of cuttings a good proxy to explore phenological responses of temperate forests in warming and photoperiod experiments? *Tree Physiology* **34**:174–183.

513 Vitasse, Y., S. Delzon, E. Dufrene, J.-Y. Pontailier, J.-M. Louvet, A. Kremer, and R. Michalet. 2009.
514 Leaf phenology sensitivity to temperature in European trees : Do within-species populations exhibit
515 similar responses? *Agricultural and Forest Meteorology* **149**:735–744.

516 Vitasse, Y., C. Signarbieux, and Y. H. Fu. 2018. Global warming leads to more uniform spring
517 phenology across elevations. *Proceedings of the National Academy of Sciences* **115**:1004–1008.

518 Webb, C. O., D. D. Ackerly, M. A. Mcpeek, and M. J. Donoghue. 2002. Phylogenies and community
519 ecology. *Annual Review of Ecology Evolution and Systematics* **33**:475–505.

520 Wiens, J. J., D. D. Ackerly, A. P. Allen, B. L. Anacker, L. B. Buckley, H. V. Cornell, E. I. Damschen,
521 T. Jonathan Davies, J.-A. Grytnes, S. P. Harrison, et al. 2010. Niche conservatism as an emerging
522 principle in ecology and conservation biology. *Ecology Letters* **13**:1310–1324.

523 Wolkovich, E. M., and E. E. Cleland. 2014. Phenological niches and the future of invaded ecosystems
524 with climate change. *Annals of Botany PLANTS* **6**:1–16.

525 Wright, I. J., P. B. Reich, M. Westoby, D. D. Ackerly, Z. Baruch, F. Bongers, J. Cavender-Bares,
526 T. Chapin, J. H. C. Cornelissen, M. Diemer, J. Flexas, E. Garnier, P. K. Groom, J. Gulias,
527 K. Hikosaka, B. B. Lamont, T. Lee, W. Lee, C. Lusk, J. J. Midgley, M.-L. Navas, Ü. Niinemets,
528 J. Oleksyn, N. Osada, H. Poorter, P. Poot, L. Prior, V. I. Pyankov, C. Roumet, S. C. Thomas, M. G.
529 Tjoelker, E. J. Veneklaas, and R. Villar. 2004. The worldwide leaf economics spectrum. *Nature*
530 **428**:821–827.

531 Yu, H., E. Luedeling, and J. Xu. 2010. Winter and spring warming result in delayed spring phenology
532 on the Tibetan Plateau. *Proceedings of the National Academy of Sciences of the United States of*
533 *America* **107**:22151–22156.

534 Zeng, Z. A., and E. M. Wolkovich. 2024. Weak evidence of provenance effects in spring phenology
535 across Europe and North America. *New Phytologist* **242**:1957–1964.

536 Zohner, C. M., and S. S. Renner. 2014. Common garden comparison of the leaf-out phenology of
537 woody species from different native climates, combined with herbarium records, forecasts long-term
538 change. *Ecology Letters* **17**:1016–1025.

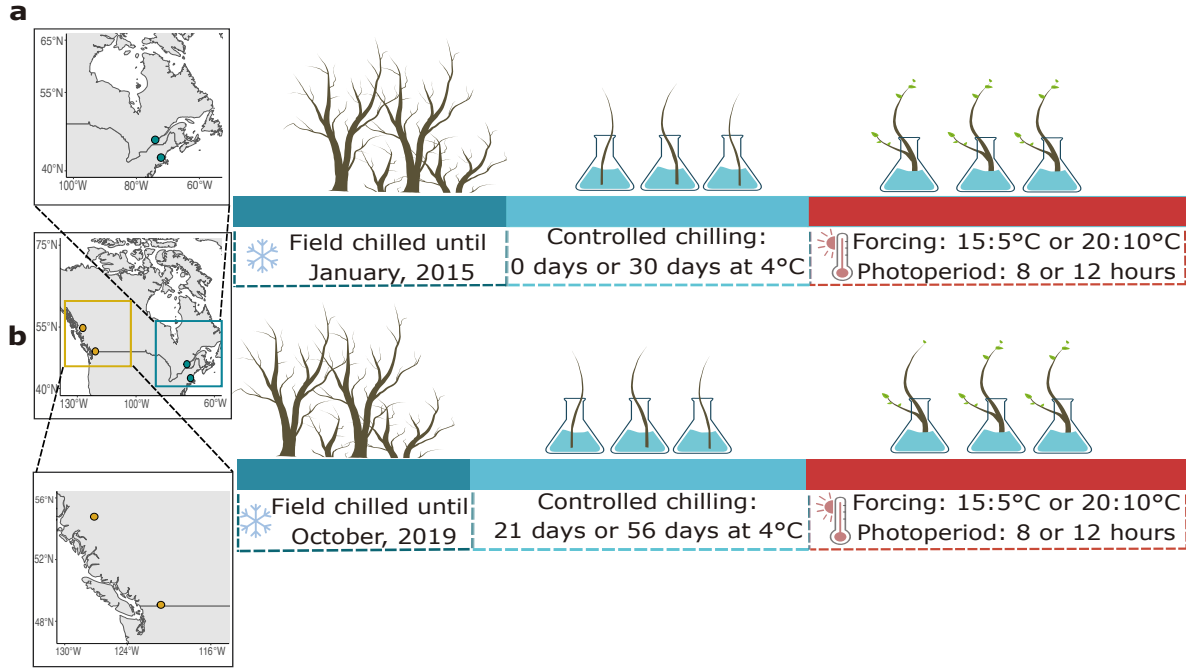
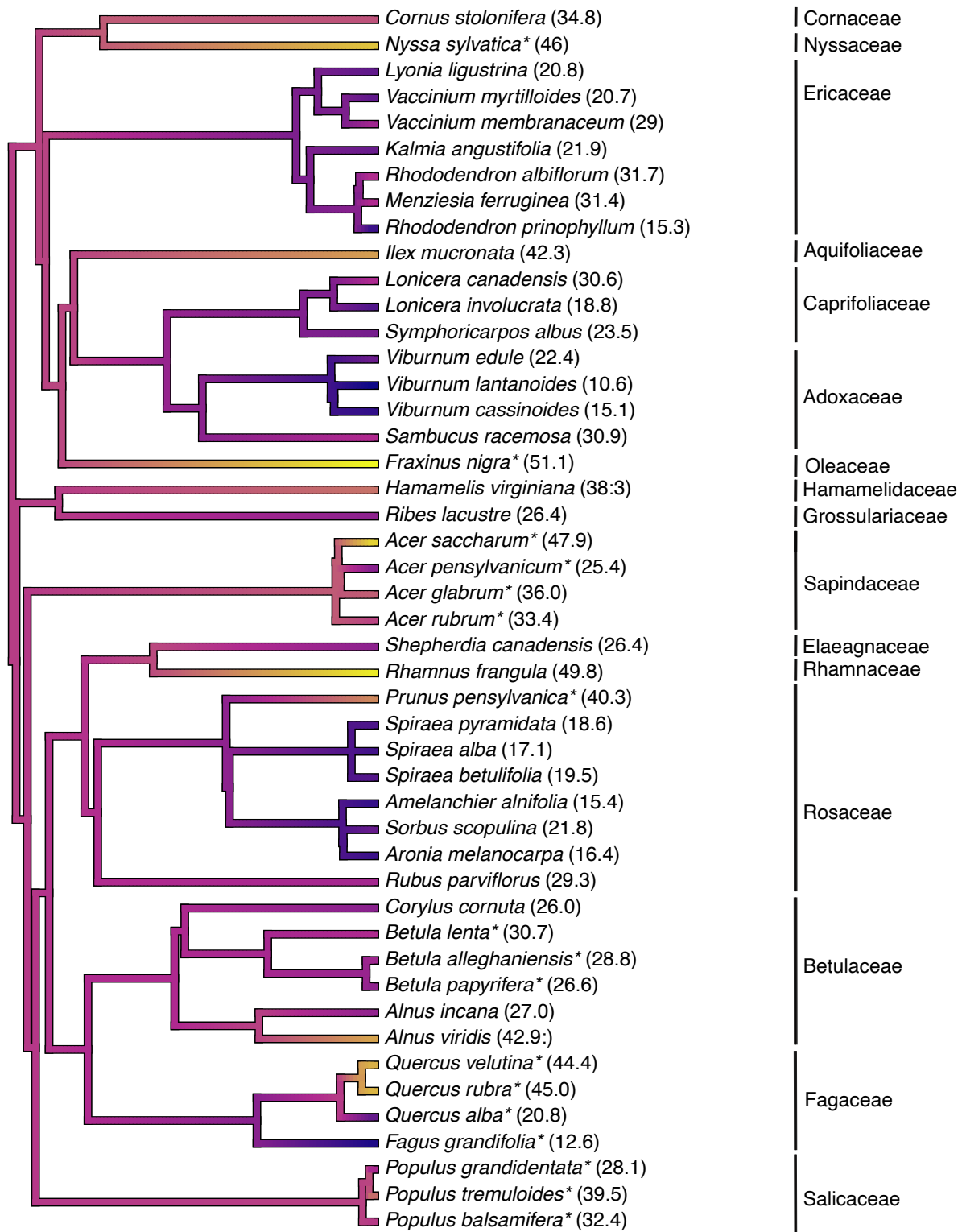


Figure 1: We combined data from two controlled environment studies of temperature and photoperiod cues for budburst in temperate woody species **a**, from two forest communities in eastern North America that span a 4° latitudinal gradient and **b**, two communities in western North America that span a 6° latitudinal gradient. Combined, our dataset includes 28 tree and shrub species from eastern communities and 21 species from western communities, with three species occurring across all four populations. Both studies used a full-factorial design with two levels each of chilling treatments (consisting of a combination of field and different duration of chamber chilling), forcing and photoperiod, for a total of eight unique treatments. Observations of budburst were made during the forcing treatment, using the BBCH scale to classify stages of bud development (see Loughnan and Wolkovich, 2024; Savas et al., 2017, for guides we made for these species).



10 Intercept (days) 50

Figure 2: Tree (denoted with an asterisk) and shrub species differences were accounted for by including phylogenetic effects on the intercept of a model estimating days to budburst after the start of forcing treatments (simplified to 'days to here'). This approach estimated a phylogenetic scaling parameter for the intercept of 0.4 (where values close to one indicate a phylogenetic effect consistent with Brownian motion and values close to zero the absence such an effect).

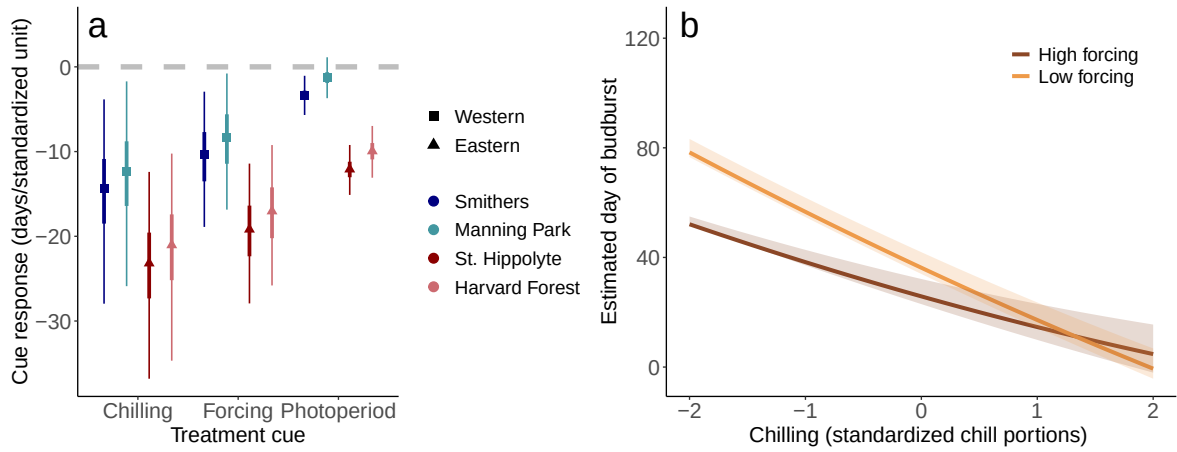


Figure 3: **a**, Posterior distributions of estimated chilling, forcing, and photoperiod responses with individual site-level effects. Symbols denote the eastern (triangle) and western (square) transects and the darker point color our more northern communities. Points represent the mean, thicker lines the 50% uncertainty interval, and the thin lines the 90% uncertainty interval. **b**, Cues interacted to produce a subadditive effect, where delays in budburst under low chilling was offset by high forcing conditions, and vice versa. To make results comparable across cues we standardized chilling, forcing, and photoperiod via z -scores using two standard deviations, see methods for further details.

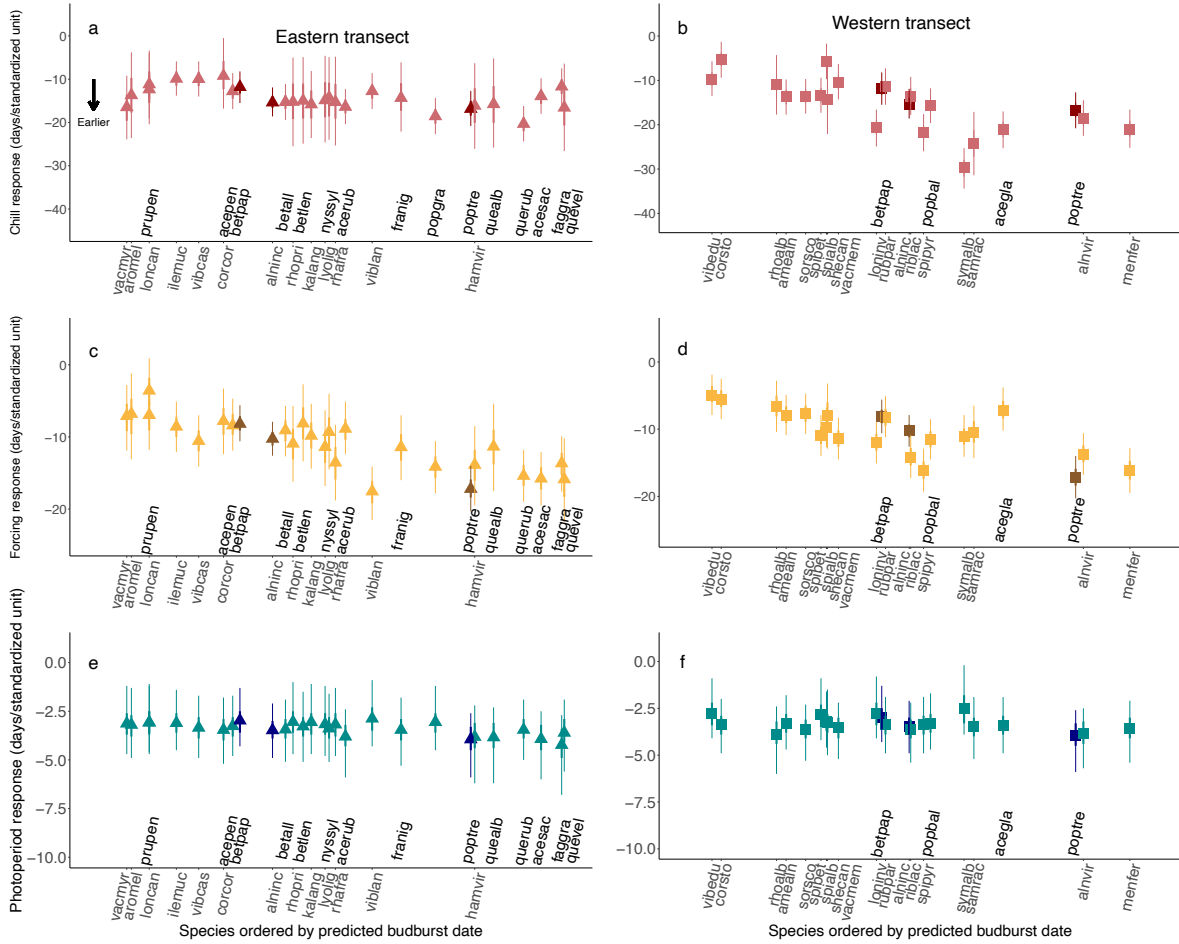


Figure 4: Estimated species **a,b** chilling, **c,d** forcing, and **e,f** photoperiod responses ordered by estimated budburst dates for both the **a,c,e** eastern and **b,d,f** western sites based on our model estimates. Cues are plotted on differing y-axis scales to better depict species differences across cues. For each species, the points represent the mean, thinner error bars represent the 90% uncertainty interval, and thicker bars represent the 50% uncertainty interval. The three species that occur in both the eastern and western transects are depicted using darker colours and tree species denoted using an asterisk following the species abbreviation. To make the results across cues directly comparable, we standardized predictors (here, chilling, forcing, and photoperiod) via z-scores using two standard deviations, see methods for further details.

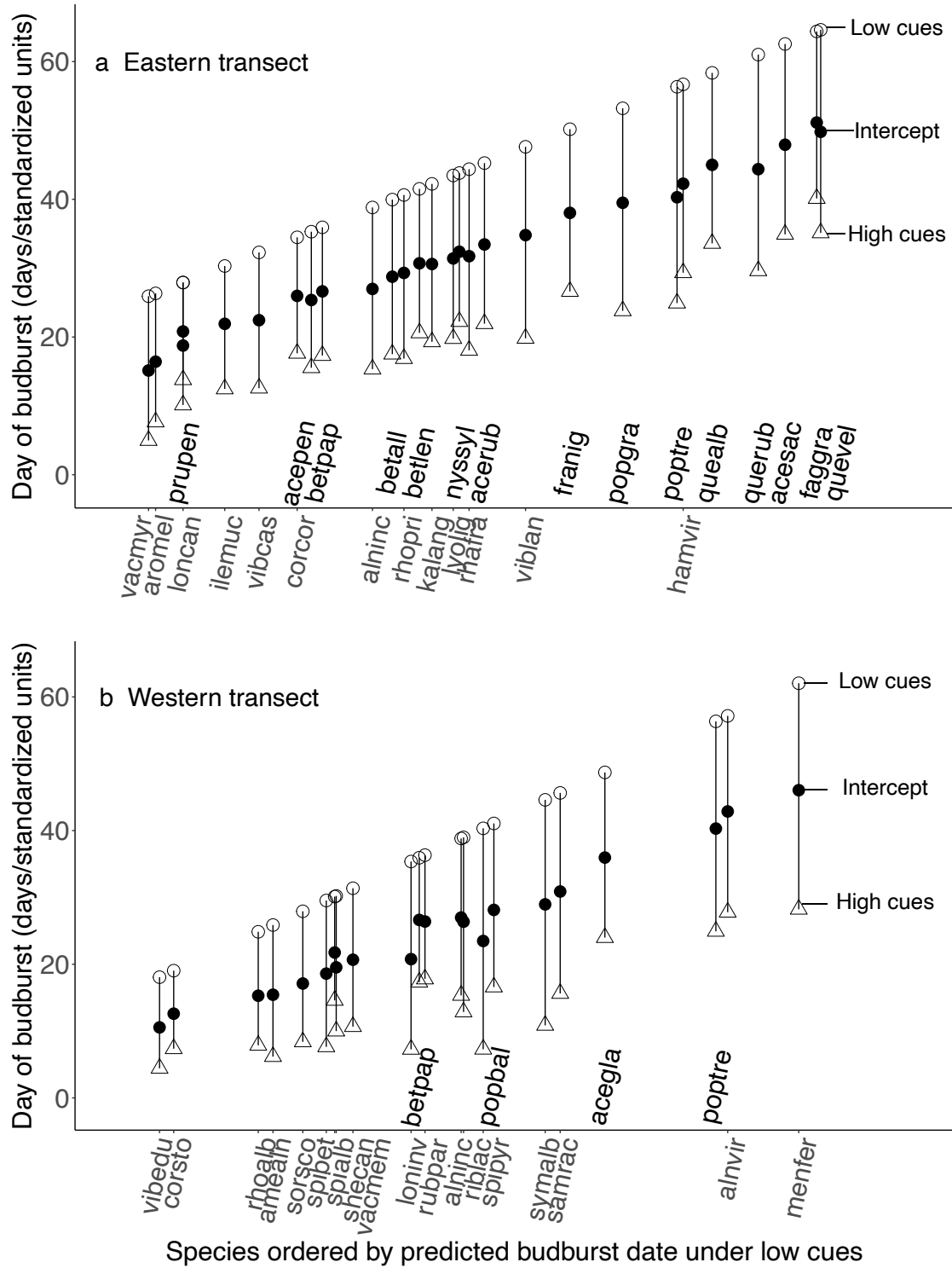


Figure 5: Comparisons of estimated day of budburst for tree (denoted with an asterisk) and shrub species based on the full model (intercept plus all cues), shown as white (open) symbols, versus the intercepts only (without effects of chilling, forcing, and photoperiod), shown as black (filled) symbols. Points are ordered according to differences in species estimated budburst dates for both the **a**, eastern **b**, and western sites, with low cue conditions, depicted as circles, and high cue conditions, depicted as triangles.